Tutorial 1A (to be completed in week1)
Database Concepts & Oracle Introduction

Oracle Clients:

Both SQL Developer and JDeveloper are GUI-based clients that will typically be used in this course for interacting with the Oracle database (e.g., 12c). You will access these clients via VMware Horizon Client* https://remote.massey.ac.nz which in general is used for accessing software from Massey Computing labs.

SQL Developer - In this course most interaction i.e., access to the database (e.g., running, editing & debugging of SQL and PL/SQL statements, etc.) would be done using SQL Developer.

The lab has SQL Developer version 21.4.3. While in you can download these clients and use them but because of security issues you cannot interact with the Oracle server. Hence you must use these clients via Massey's General Labs image.

JDeveloper - mainly used for drawing database diagrams but could also be used for some interaction with the Oracle database.

SQL Developer Data Modeler – Another GUI-based client for data modelling tasks (e.g., create, browse, and edit logical, relational, physical data models). Data Modeler can be invoked from the SQL Developer menu (Tools→Data Modeler). However, this is also available as a standalone application which is also installed in the lab. If interested feel free to explore this in your own time. https://www.oracle.com/database/technologies/appdev/datamodeler.html points to various resources that you might find helpful.

Tutorial Objectives:

a. Test your Oracle account using SQL Developer
b. Run Oracle script file(s) in SQL Developer
c. Learn some important database concepts e.g., file structure (field, record, etc.), data redundancy (detection, anomalies, elimination, etc.)

*Access is allowed only through VMware because of security reasons. This is ITS decision that is binding on us. This means, you do not need to download and install these software client applications (SQL Developer, JDeveloper, etc). You need to use your Massey username and password to access s/w via VMWare.

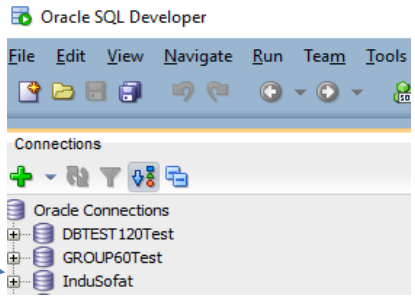
Part A

a. Test your Oracle a/c using SQL Developer

Log in to your Oracle account using SQL Developer. The login details {username (e.g., IT337001) and password} are provided in the file – Oracle Usernames and Passwords – posted on the Stream site. Allocated username and password are listed against your Student ID. You must only use your allocated username.

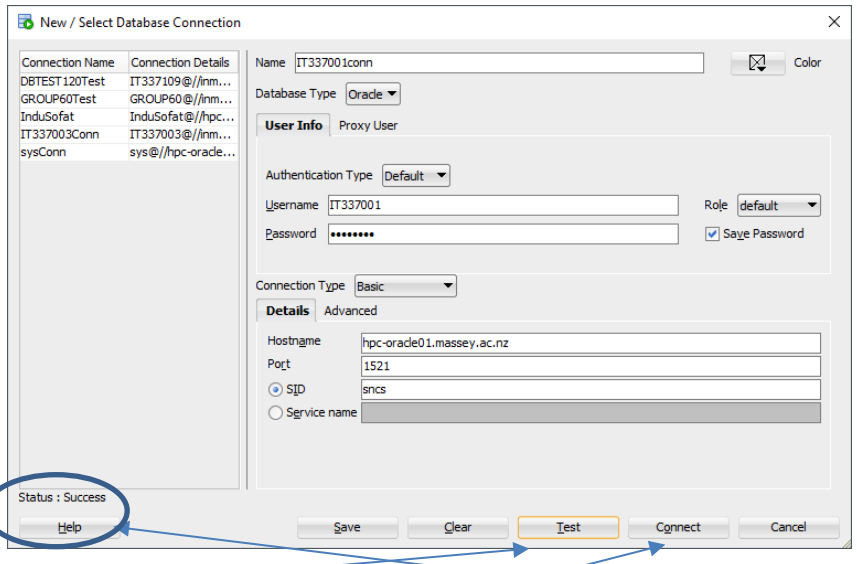
i. Open SQL Developer (Start→ Oracle→SQL Developer); alternatively, search directly for the application SQL Developer.

Click the Green plus tab for a New Connection to get the New/Select Database Connection window as shown.



Ignore the other pre-existing connections here in the image.

Fill in the following details as provided with ***your*** allocated Oracle user account (and **not** IT337001).
Hostname (server for Oracle database): **hpc-oracle01.massey.ac.nz**
(Listener) Port: **1521**
SID: **sncs**



ii. Test your connection before connecting to Oracle server. The Status should change to Success or Failure (if the test is un-successful for some reason). Click the Connect button when Status shows Success.

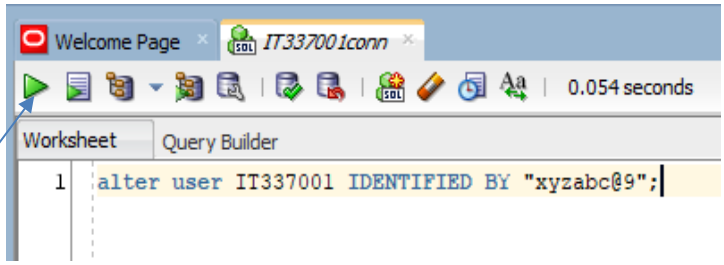
iii. Next *change the password from the given default.*
Use the following SQL (Structured Query Language) statement (replace the values inside the brackets <>):

alter user <username> identified by <"new password">;

Example: User IT337001 will issue the following statement to change to new password xyzabc@9 from the given default password IT337001:

alter user IT337001 IDENTIFIED BY "xyzabc@9";

Note: Do not use <> in your password change statement, i.e., just replace <username> with your allocated username (e.g., IT337XXX **without** <>) while changing the password. Also, use the **quotation marks** around the changed password. ***Make a note of this new password.***



Put a semi-colon (;) at the end of every SQL statement to indicate that the statement has ended.

Press the Green arrow (Ctrl+Enter) to Run (a SQL) Statement.

iv. Exit SQL Developer and log in again using your new password.

NOTE: It is your responsibility to change the password from the given default and secure your Oracle account. Please keep your password secured, as you will be using this account throughout the semester for tutorial exercises and marked assessments.

Part B

b. Working with Oracle script file

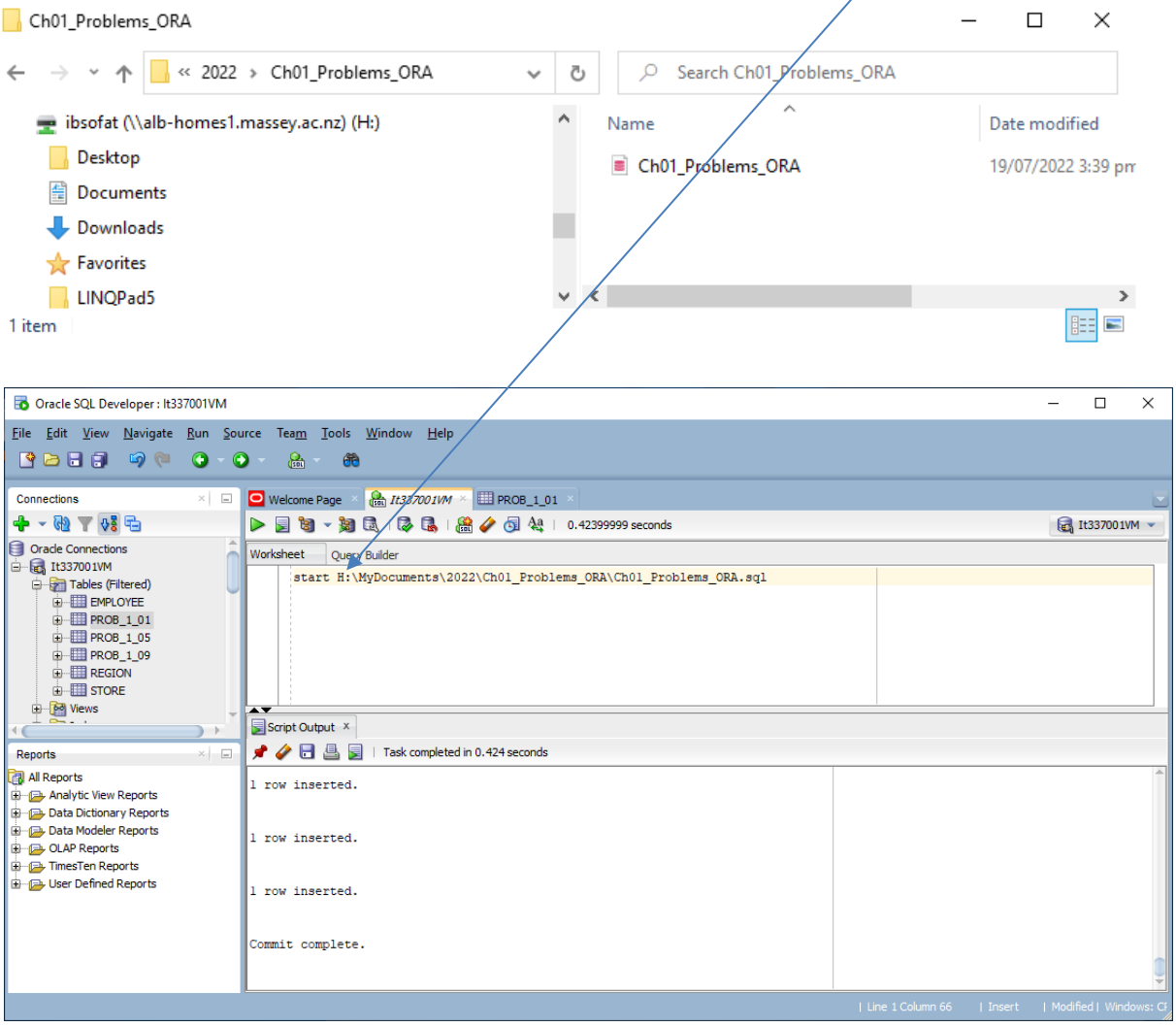
Download Zipped Oracle Script file from the Stream site. Unzip this to get the code file - Ch01_Problems_ORA.sql – on your local machine. This script file will be used for first creating three tables and then populating (load) data into these.

i. Run the script file - you need to say **start** or **@** followed by your file path and the file name. e.g.

start c:\Ch01_Problems_ORA; --assuming the file is on c:*
or
@ c:\Ch01_Problems_ORA; --assuming the file is on c:
or
@ c:\Ch01_Problems_ORA.sql; --assuming the file is on c:

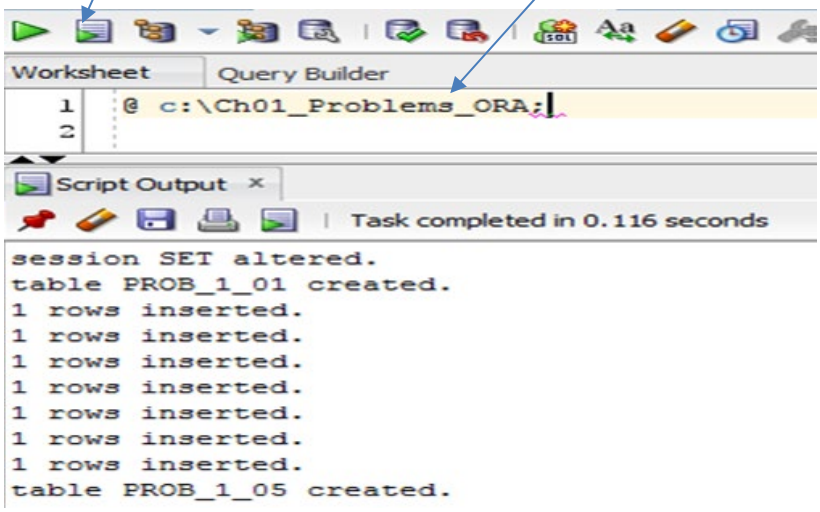
----updated after publishing for VMware related requirements to run a script file

* With use of VMWare to access the application and server it would be your H: drive where you can save the files (as in the image below). One can shuttle between One Drive and H: drive once VMware is running.
Also remember to un-compress the provided folder and the file path needs to have the folder name before the file name if your file is part of the folder, otherwise it is not needed.

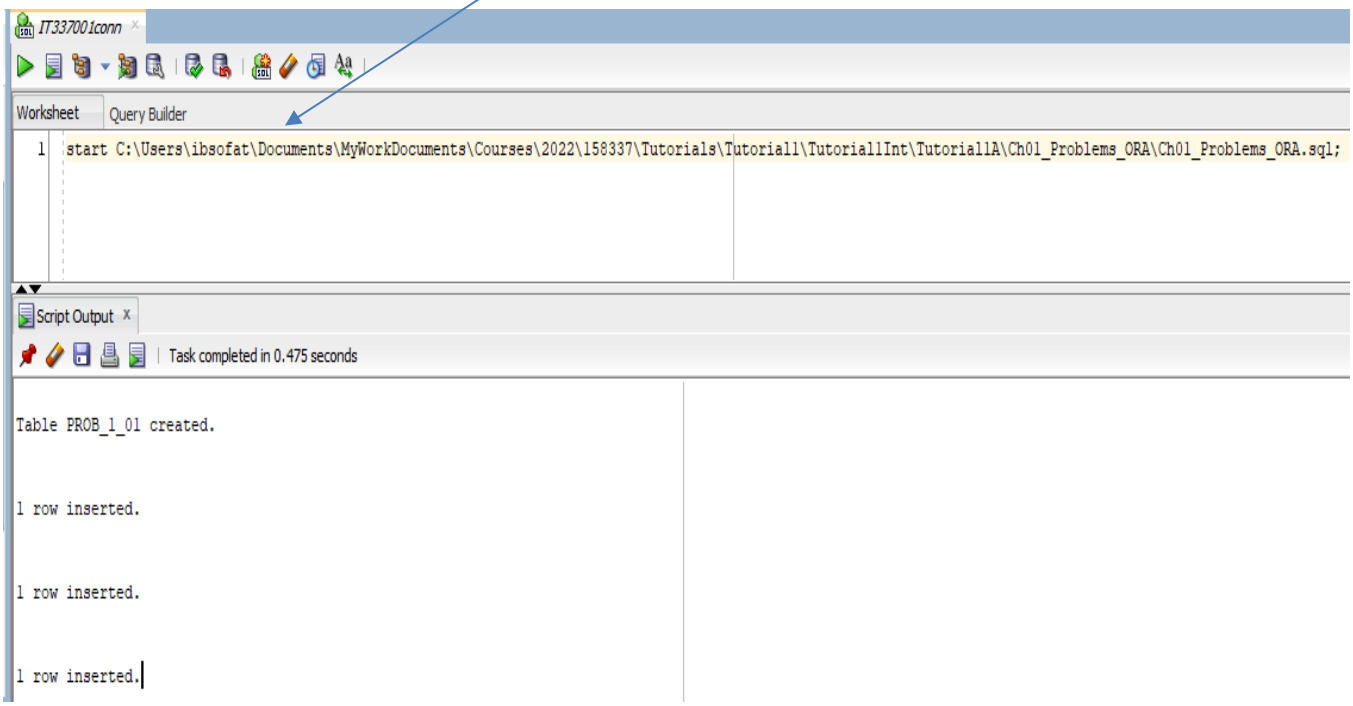


Click the Green arrow to Run the script file (here the file is not under any folder).

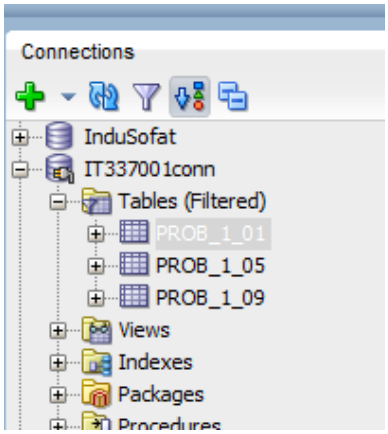
Although the script file has an extension .sql, it is not necessary to mention .sql while running the script file. The file path should not contain any spaces or wild characters.



Here is another example where the .sql file is placed under many folders (no wild characters or spaces in a long path).



Expand various nodes (e.g., Tables) to look for the objects under these. You will probably see only three tables at this stage (as in the figure on right).



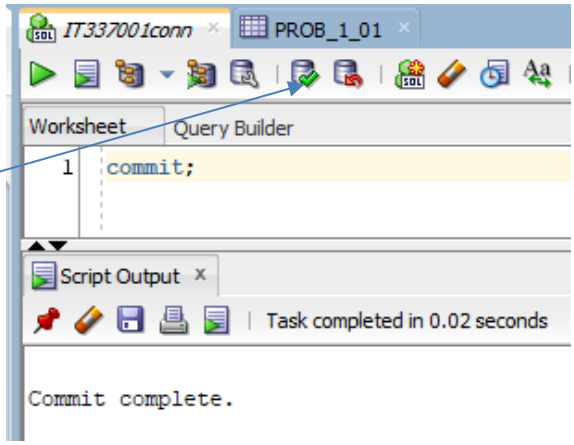
ii. You could also see the list of tables that exist in your Oracle a/c, by issuing the following SQL statement:

select table_name from user_tables;

iii. To store the inserted data permanently in the table(s) or make any other changes permanent, issue the commit statement (another SQL statement):

commit;

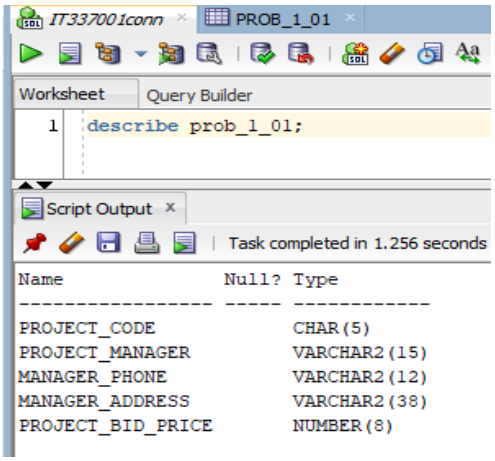
Or click Commit button.



iv. To see the table structure, issue the following SQL Plus command:

describe <tablename>

(Without or with semi-colon; first 4 letters should also suffice)

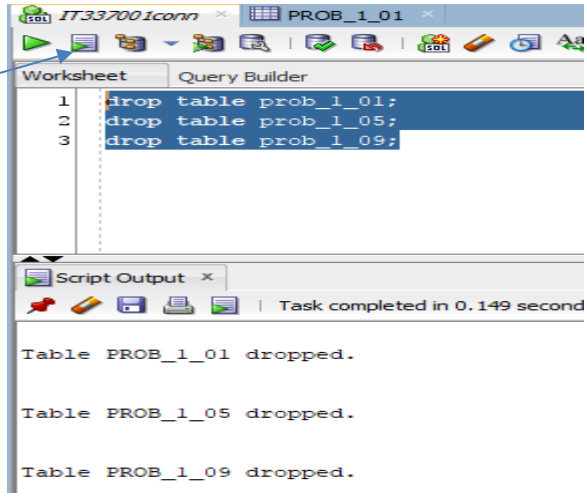


You can also double click on a Table Name (e.g., PROB_1_01) to see its various details.
The Columns tab (default) shows column details such as Column Name, Data Type, Null(?), etc.

v. To drop* a table, issue the following SQL statement:

drop table <tablename>;

Click Run Script (if multiple code lines are there)



Note*: Use the drop statement only when you do not need a table and its data. However, if you drop any of the tables that are required later then you can recreate these (save your script files for such situations so that you can rebuild your database).

Note: Commands themselves are **not** case-sensitive; however, it is a good practice to follow a consistent case formatting style for enhanced readability.

Part C

c. Learn some Database concepts

i. Repeat Part b step 'i.' to create the three tables and then populate them with data.

ii. Answer the following eight problems with the given file structure of tables.

Use Fig P1.1, pg 32 to answer the following Problems 1-4.**

1. How many records does the file contain? How many fields are there per record?
2. What problem would you encounter if you wanted to produce a listing by city? How would you solve this problem by altering the file structure?
3. If you wanted to produce a listing of the file contents by last name, area code, city, state, or zip code, how would you alter the file structure?
4. What data redundancies do you detect? How could those redundancies lead to anomalies?

Use Fig P1.5, pg 33 to answer the following Problems 5-8.

5. Identify and discuss the serious data redundancy problems exhibited by the file structure shown in Figure P1.5.
6. Looking at the EMP_NAME and EMP_PHONE contents in Figure P1.5, what change(s) would you recommend?
7. Identify the various data sources in the file you examined in Problem 5.
8. Given your answer to Problem 7, what new files should you create to help eliminate the data redundancies found in the file shown in Figure P1.5?

Screen dump of the tables needed for the above problems

FIGURE P1.1 The File Structure for Problems 1-4

PROJECT_CODE	PROJECT_MANAGER	MANAGER_PHONE	MANAGER_ADDRESS	PROJECT_BID_PRICE
21-5Z	Holly B. Parker	904-338-3416	3334 Lee Rd., Gainesville, FL 37123	16833460.00
25-2D	Jane D. Grant	615-898-9909	218 Clark Blvd., Nashville, TN 36362	12500000.00
25-5A	George F. Dorts	615-227-1245	124 River Dr., Franklin, TN 29185	32512420.00
25-9T	Holly B. Parker	904-338-3416	3334 Lee Rd., Gainesville, FL 37123	21563234.00
27-4Q	George F. Dorts	615-227-1245	124 River Dr., Franklin, TN 29185	10314545.00
29-2D	Holly B. Parker	904-338-3416	3334 Lee Rd., Gainesville, FL 37123	25559999.00
31-7P	William K. Moor	904-445-2719	216 Morton Rd., Stetson, FL 30155	56850000.00

FIGURE P1.5 The File Structure for Problems 5-8

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CODE	JOB_CHG_HOUR	PROJ_HOURS	EMP_PHONE
1	Hurricane	101	John D. Newson	EE	85.00	13.3	653-234-3245
1	Hurricane	105	David F. Schwann	CT	60.00	16.2	653-234-1123
1	Hurricane	110	Anne R. Ramoras	CT	60.00	14.3	615-233-5568
2	Coast	101	John D. Newson	EE	85.00	19.8	653-234-3254
2	Coast	108	June H. Settlemeir	EE	85.00	17.5	905-554-7812
3	Satellite	110	Anne R. Ramoras	CT	62.00	11.6	615-233-5568
3	Satellite	105	David F. Schwann	CT	26.00	23.4	653-234-1123
3	Satelite	123	Mary D. Chen	EE	85.00	19.1	615-233-5432
3	Satellite	112	Allecia R. Smith	BE	85.00	20.7	615-678-6879

Final Notes:

** pg in our tutorial documents will refer to your course textbook (13th edition) page number(s).

Textbook for 158.337: *Database Systems: Design, Implementation, and Management,* by Carlos Coronel; Steven Morris; **(13th Edition)**

If you have an older version, then it is your responsibility to look for the equivalent page numbers.